PQC DAILY STUDY SERIES

Understanding NIST's PQC Security Levels

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Understanding NIST's PQC Security Levels

As we transition to Post-Quantum Cryptography (PQC), it's crucial to understand not just the algorithms themselves, but also how their strength is measured. The National Institute of Standards and Technology (NIST) has defined a set of "security strength categories" to standardize the evaluation of PQC algorithms. These levels are designed to be at least as hard to break as well-established classical cryptographic algorithms.

The security of these new standards is benchmarked against the computational difficulty of breaking current cryptographic algorithms using both classical and quantum computers. This ensures that a "Level 1" PQC algorithm is at least as secure as AES-128, and so on. Understanding these levels is key for selecting the right algorithm for a given application, balancing security with performance.

The Five Security Levels Explained

NIST has defined five distinct security strength levels, each corresponding to a classical cryptographic counterpart. This allows for a clear comparison and risk assessment when migrating systems.

- **Level 1:** At least as hard to break as AES-128. This level is suitable for many applications, providing a baseline of quantum resistance.
- **Level 2:** At least as hard to break as SHA-256/256-bit elliptic curve cryptography. A significant step up in security.
- **Level 3:** At least as hard to break as AES-192. Recommended for systems requiring higher security margins.
- **Level 4:** At least as hard to break as SHA-384/384-bit elliptic curve cryptography.
- **Level 5:** At least as hard to break as AES-256. This is the highest level, intended for applications protecting the most sensitive data with a very long-term security horizon.

> **Applicability to Internet Protocols**
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> In protocols like TLS 1.3, different security levels can be negotiated. For instance, a client and server might negotiate a Level 1 ML-KEM for session key establishment, providing a

> good balance of performance and security. For digital signatures in software updates (a long-term security need), a higher level like ML-DSA at Level 3 or 5 might be more appropriate. The choice depends on the data's sensitivity and the performance requirements of the protocol.

Comparison of NIST PQC Standards by Security Level

Algorithm	Type	Available Security Levels	Primary Use Case
ML-KEM (Kyber)	Key Encapsulation Mechanism	1, 3, 5	Key exchange in TLS, VPNs
ML-DSA (Dilithium)	Digital Signature Algorithm	2, 3, 5	Digital signatures, code signing
SLH-DSA (SPHINCS+)	Digital Signature Algorithm	1, 3, 5	Digital signatures (stateless)

Daily Quiz

1. What classical algorithm's security is NIST PQC Security Level 1 benchmarked against?
2. Which of the standardized PQC algorithms is a Key Encapsulation Mechanism (KEM)?
3. For which use case might a higher security level (e.g., Level 5) be particularly important?